01 Foundational Concepts of Analytics

1. BUSINESS ANALYTICS

Business Analytics is about using data to help business make better decisions. It combines data, tools and thinking to solve real business problems and improve performance.

Example: A store looks at past sales data to decide which products to stock more of during busy seasons.

Source: (Investopedia, 2024)

2. DATA ANALYTICS

Data analytics focuses on analysing raw data to find patterns and insights. It is more about working with the data itself, while business analytics focuses on using those insights to make decisions.

Example: A data analyst studies customer data to find trends in buying behaviour.

Source: (IBM, 2023)

3. BUSINESS INTELLIGENCE (BI)

Business Intelligence is about using tools like dashboards and reports to see what is currently happening in a business. Mainly looks at past and present data rather than predicting the future.

Example: A manager uses a dashboard to track daily sales across different stores.

Source: (Microsoft, 2024)

4. Data-driven decision making
This means making decisions based on data instead of guessing or relying only on experience. It helps businesses be more accurate and objective.

Example: A company choose the best marketing campaign based on data showing which one preformed better.

Source: (Havard Business Review, 2020)

5. BIG DATA
Refers to very large and complex sets of data that are difficult to handle with normal tools. It usually involves huge amounts of fast-moving and different data types.

Example: A platform like Netflix uses massive amounts of user data to recommend shows.

Source: (Oracle: 2023)

6. DESCRIPTIVE ANALYTICS
Descriptive analytics explains what has already happened in the past using data. It usually involves reports, summaries and basic charts.

Example: A report showing how much a company sold the previous month.

Source: (Investopedia, 2024)

7. DIAGNOSTIC ANALYTICS

~~Predictive analytics~~ looks at data to understand why something happened. It tries to find the cause behind a trend or problem.

Example: A business finds out sales dropped because prices were increased.

Source: (IBM, 2023)

8. PREDICTIVE ANALYTICS

Uses data to make predictions about what might happen in the future

Example: An online store predicts which customers are likely to buy again soon.

Source: (SAS, 2022)

9. PRESCRIPTIVE ANALYTICS

This goes a step further by suggesting what action should be taken based on data. It helps business decide the best course of action

Example: A delivery company gets route suggestions to save time and fuel.

Source: (Oracle, 2023)

11. DATA LITERACY

It is the ability to understand and work with data. It

means being able to read charts, interpret results and use data to make decisions.
Example: A team leader reads a performance dashboard and uses it to improve team results.

Source: (Gartner, 2021)

Reference List

* Investopedia. (2024). Business analytics and descriptive Analytics
* IBM. (2023) What is Data Analytics? https://www.ibm.com
* Harvard Business Review. (2020) - The Value of Data-driven decision Making. https://hbr.org
* Oracle. (2023) Big data & Prescriptive Analytics? www.oracle.com
* Gartner. (2021) Data Literacy definition. https://www.gartner.com

02 DATA, MEASUREMENT & REPORTING

11. STRUCTURED DATA

Data that is neatly organized into rows and columns, usually in tables. It is easy to store and analyse (IBM, 2023)

Example: A company stores customer details like name, age and purchase history in a table.

12. UNSTRUCTURED DATA

Data that does not follow a fixed format, such as text, images or videos. It is harder to organise and analyse unstructured data (IBM, 2023)

Example : Customer reviews, social media posts.

13. DATA WAREHOUSE

A central ~~search~~ place where large amounts of data from different sources are stored for analysts. Helps businesses access clean and organised data in one location. (Oracle, 2023)

Example : A retail company combines sales, customer, and inventory data in one system for reporting

14. DATA MINING

The process of analysing large datasets to find patterns, trends or useful information. Helps business discover insights they may not easily identify (Investopedia, 2024)

Example : A supermarket discovers that customers who buy bread often also buy milk

15. DATA VISUALIZATION

The use of charts, graphs and visuals to present data in a clear and understandable manner. (Tableau, 2023)

Example : A bar chart showing monthly sales performance across different regions.

16. DASHBOARD

A visual display that shows key data and metrics in one place, often updated in real time. Helps user quickly monitor performance (Microsoft, 2024)

Example : Dashboard showing daily revenue, targets, top-performing products

17. METRIC

Any measurable value that helps track performance or activity in a business. Not all metrics are important for decision-making (Investopedia, 2024)

Example: The number of website visitors per day is a metric.

18. KEY PERFORMANCE INDICATOR (KPI)

A specific type of metric that is directly linked to business goals and used to measure success. While all KPIs are metrics, not all metrics are KPIs (Harvard Business Review, 2020)

Example: For an online store, conversion rate may be a KPI while total page views may just be a metric.

19. BENCHMARK

A standard or reference point used to compare performance. It helps businesses understand how well they are doing compared to the standard. (Investopedia, 2024)

Example: A company compares its sales growth to the industry average.

20. SEGMENTATION

The process of dividing customers into groups/segments based on shared characteristics like age, behavior or preferences. Helps target groups effectively (Hubspot, 2023)

Example: A company separates customers into groups like students, professionals and retirees for marketing.

03 CUSTOMER & SALES KPIs

21 CONVERSION RATE

The percentage of people who take a ~~desired~~ desired action out of the total number of visitors. It shows how effective a business is at turning interest into actual results (Hubspot, 2023)

Example: If 100 people visit a website and 5 make a purchase, the conversion rate is 5%.

22 CUSTOMER ACQUISITION COST (CAC)

The total cost a business spends to gain one new customer. This includes marketing, advertising and sales expenses (Investopedia, 2024)

Example: If a company spends R10 000 on marketing and gains 100 customers, the CAC is R100 per customer.

23 CUSTOMER LIFETIME VALUE (CLV/LTV)

The total amount of revenue or profit a business expects to earn from a customer over the entire relationship. Measures how valuable a customer is over time (HubSpot, 2023)

Example: If a customer spends R500 every year for 5 years, their CLV is R2500.

24 CLV : CAC Ratio

This ratio compares how much value a customer brings versus how much it costs to acquire them. A healthy business usually has a ratio where CLV is much higher than CAC (e.g. 3:1)

24. CHURN RATE

The percentage of customers who stop doing business with a company over a period of time. Shows how many customers are being lost (Investopedia, 2024)

Example: If a company keeps 90 out of 100 customers, the churn ~~retention~~ rate is ~~90%~~ 10%.

25. CUSTOMER RETENTION RATE

The percentage of customers a business keeps over a certain period. It is the opposite of churn and shows customer loyalty (Hubspot, 2023)

Example: If a company keeps 90 out 100 customers, the retention rate is 90%

26. NET PROMOTER SCORE (NPS)

Measures how likely customers are to recommend a business to others. It is based on a simple survey asking customers to rate their likelihood to recommend. (Bain & Company, 2022)

Example: Customers rate a company 0-10 and the business calculates a score based on promoters and detractors.

27. AVERAGE ORDER VALUE

The average amount of money customers spend per transaction. Helps understand customer spending behaviour (Shopify, 2023)

Example: Total revenue is R5000 from 100 orders, then AOV is R50

28. SALES FUNNEL

Describes the journey customers go through from first hearing about a product to making a purchase. It shows how potential customers move through different stages

Example: A customer sees an ad (awareness), visits website (interest) and then buys product (conversion).

04 MARKETING & CAMPAIGN ANALYTICS

29. MARKETING CAMPAIGN

A planned set of activities designed to promote a product or service. Usually runs across different channels like social media, emails or ads

Example: A company runs a month-long Instagram and email campaign to promote a new product launch

30 POST-CAMPAIGN ANALYSIS

The process of reviewing a campaign after it ends to see what worked and what didn't to improve future campaigns (Google Analytics, 2023)

Example: A business checks which ads brought in the most sales after a campaign ends

31. INCREMENTAL ~~VALUE~~ REVENUE

The additional revenue generated as a direct result of a marketing campaign. Shows the true impact of the campaign (Investopedia, 2024)

Example: If sales were usually R50000 but increased to R70 000 during a campaign, incremental revenue is R20 000.

32. RETURN ON INVESTMENT (ROI)

Measures how much profit is made compared to the cost of investment. Used to evaluate whether a campaign or activity was worth it (Investopedia, 2024)

Example: If a campaign costs R10 000 and generates R20000 in profit, the ROI is positive.

33. RETURN ON AD SPEND (ROAS)

Measures how much revenue is earned for every rand spent on advertising. Focuses solely on ad performance (Google Ads, 2023)

Example: If a business spends R1000 on ads and earns R4000 in revenue, The ROAS is 4:1.

34. CLICK-THROUGH RATE (CTR)

The percentage of people who click on an ad or link after seeing it. Shows how engaging or relevant the content is (Google Ads, 2023)

Example: If 1000 people ~~seen~~ see an ad and 50 click on it, the CTR is 5%.

35. BOUNCE RATE

The percentage of visitors who leave a website after viewing only one page. A high bounce rate suggests an irrelevant or not engaging page (Google Analytics, 2023)

Example: Most users leave a website without clicking anything else, the bounce rate is high

36. A/B TESTING

A method of comparing two versions of something to see which one performs better. Commonly used in marketing to improve results (HubSpot, 2023)

Example: A company tests two different email subject lines to see which one gets more opens

37. LEAD GENERATION

The process of attracting and capturing potential customers who show interest in a product or service (HubSpot, 2023)

Example: A company collects email addresses through a sign-up form on its website

38. MARKETING ATTRIBUTION

The process of identifying which marketing channel contributed to a sale or conversion. Helps understand what is driving results (Google Analytics, 2023)

Example - A company finds that most ~~values~~ sales come from Instagram rather than email marketing